

Jieun Kim

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RESEARCH INTERESTS

Trustworthy Machine Learning, Model Reliability & Robustness, Representation Learning, Brain-Inspired AI

EDUCATION

Korea University Mar 2021 – Feb 2023

M.S. in Industrial and Management Engineering

Advisor: Jun-Geol Baek

Sookmyung Women's University Mar 2017 – Feb 2021

B.S. in Statistics and Computer Science

Advisor: Young-Geun Choi

RESEARCH EXPERIENCE

Korea Institute of Science and Technology (KIST) May 2024 – Feb 2026

Research Intern (Advisor: Taegon Kim)

- Constructed large-scale spiking neuron networks (SNN) to quantitatively simulate and evaluate nonlinear dynamics of motor and non-motor symptoms in Parkinson's disease.
- Investigated neural synchronization and network-level dynamics to optimize underlying predictive frameworks and metrics for adaptive deep brain stimulation.

Georgia Institute of Technology, USA Aug 2023 – Feb 2024

Visiting Researcher (Advisor: Jungho Lew)

- Engineered deep generative adversarial networks (GANs) and temporal architectures to detect multidimensional anomaly spikes in high-frequency building sensor time-series data.

Korea University Mar 2021 – Feb 2023

Graduate Researcher (Advisor: Jun-Geol Baek)

- Developed a multiple encoder framework for multivariate time-series anomaly detection under varying normality conditions (resulting in a manuscript as the second author).
- Formulated robust predictive models for Remaining Useful Life (RUL) estimation leveraging high-dimensional vehicle telemetry and operational logs.
- Executed data-driven causal inference and statistical analysis to pinpoint critical failure variables and systemic operational vulnerabilities.

Sookmyung Women's University Jul 2019 – Jul 2020

Research Intern (Advisor: Heejoon Chae — Joint Research with Seoul National University Hospital)

- Classified heterogeneous breast cancer subtypes by training multi-layer perceptron (MLP) architectures on high-throughput multi-omics gene expression data.

Sookmyung Women's University Jun 2020 – Jul 2020

Research Intern (Advisor: Yangjin Kim)

- Performed parametric survival modeling and volatility analysis on large-scale COVID-19 clinical patient records to estimate epidemiological risk factors.

PUBLICATIONS

J. Kim, Jun-Geol Baek. A distribution-based transformer framework for time series missing value imputation, Master's Thesis, Korea University, Department of Industrial Management and

Engineering, 2023.

J. Ki, **J. Kim**, S. Kim, J. Park, J. Lee, and Y.-J. Kim. Statistical analysis of estimating incubation period distribution and case fatality rate of COVID-19, The Korean Journal of Applied Statistics, 2020.

Joungmin Choi, Jiyoung Lee, **Jieun Kim**, Jihyun Kim, Heejoon Chae. Breast Cancer Subtype Classification Using Multi-omics Data Integration Based on Neural Network, Journal of KIISE, 2020.

PRESENTATIONS

Topographic Cortico-Basal Ganglia Computational Model Reveals Motor and Non-motor Symptom Comorbidity in Parkinson's disease, Society for Neuroscience (SfN), 2025.

Computational Cortico-Basal Ganglia Model Reveals Altered Spiking Dynamics and Synchronization in Parkinson's disease, KSBNS, 2025.

Spiking Neural Network Modeling of Basal Ganglia for Parkinson's disease Prognosis and Adaptive DBS, KSBNS, 2024.

Fuel Cell Vehicle Fuel Efficiency Prediction via Data Refinement and Adversarial Generation, Korean Institute of Industrial Engineers, 2022.

TEACHING AND PROFESSIONAL SERVICE

Corporate Instructor & Programming Assistant Spring 2022 – Dec 2023

- Developed and delivered advanced machine learning and programming curricula for major enterprise partners, including **Samsung Electronics**, LG Energy Solution, LG Electronics, SK Hynix, and Hyundai Steel.

Teaching Assistant – Korea University Mar 2022 – Dec 2022

- IDS505: Data Science & AI Programming (Spring 2022)
- IDS508: Principles and Applications of Machine Learning (Fall 2022)

Academic Service 2023

- Journal Reviewer: *Journal of Imaging Science and Technology* (2023)
- Book Reviewer: *Insights from Kaggle Medalists* (2023)

AWARDS AND HONORS

Special Prize, Weather Big Data Contest 2022

Grand Prize, Data-based Service Contest for Workplace Safety AI 2021

Third Prize, Contest for Natural and Artificially Generated Data 2021

Best Paper Award, Korean Institute of Information Scientists and Engineers (KIISE) 2020

TECHNICAL SKILLS

Programming Languages: Python (PyTorch, TensorFlow, Brian2), R

Tools & Frameworks: Git, $\text{\LaTeX}$, Data Visualization Dashboards (Tableau/Matplotlib)